

ME 619: Experiments in Materials Science

Course Code	ME 619
Course Name	Experiments in Materials Science
L-T-P-C	..-.-.-3
Intended for	B.Tech/MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Module I**
Principles of Stealth Technology – camouflage, conceal, deceive, active vs. passive detection. [2 Lectures]
- **Module II**
Introduction to Signatures – radar (RCS & its reduction), infrared, visual, aural. [3 Lectures]
- **Module III** Fabrication of various materials (metals, alloys, ceramics and composites) in various forms such as single crystals, thin films, and bulk materials using physical/chemical methods. Their structural and physical properties characterization using structural characterization (Diffraction, optical and electron microscopy), Thermal characterization (DTA/DSC/TGA) and miscellaneous materials characterization tools. Scanning probe microscopy, Scanning electron microscopy, Transmission electron microscopy and X-ray diffraction etc. Physical properties measurements such as VSM, Magnetoresistance, SQUID, impedance analysis, PES, IPES, X-ray absorption spectroscopy, AFM, STEM, P-E loop, piezoelectric measurements, thermoelectric measurements etc.

Text and Reference Books

1. Kingery W. D., Bowen, H. K., Uhlhmen D. R., **Introduction to Ceramics**, 2nd Edition, John Wiley, 1976.
2. J. Reed, **Principles of Ceramic Processing**, 2nd Edition, John Wiley and sons,
3. C. R. Brundle, C.A. Evans and S.Wilson, **Encyclopedia of Materials Characterization**, Butterworth-Heinemann, 1992.
4. A.R West, **Solid State Chemistry**, Wiley
5. B.D. Cullity, **Elements of X-Ray Diffraction**, Prentice Hall, 2001.

References

1. Mahulikar, S.P., Sonawane, H.R., & Rao, G.A., 2007, **Infrared signature studies of aerospace vehicles**, *Progress in Aerospace Sciences*, Vol. 43, Nos. 7-8, pp. 218-245.
2. Rao, G.A., & Mahulikar, S.P., 2002 (Dec), **Integrated review of stealth technology and its role in airpower**, *Aeronautical Journal*, Vol. 106, No. 1066, pp. 629-641.
3. Howe D., **Introduction to the basic technology of stealth aircraft: Part 1 - basic considerations and aircraft self emitted signals (passive considerations)**, *ASME Journal of Engineering for Gas Turbines & Power*, 1991, Vol. 113, No. 75, pp. 75-79.